

# Instagram User Analytics

A SQL-driven analysis of user behavior, engagement patterns, and platform health — delivering actionable insights for marketing, product, and investor teams.



# What We Set Out to Solve

Using MySQL Workbench and a real Instagram dataset, this project translates raw user data into business intelligence across three audiences.

## Marketing Team

Identify loyal users, inactive accounts, top hashtags, and optimal ad timing to sharpen campaigns.

## Product Team

Understand engagement baselines and user behavior to inform feature prioritization.

## Investors

Measure platform activity, detect fake accounts, and assess growth signals.

# Tech Stack

## MySQL Workbench

Primary environment for querying and exploring the Instagram database.

## SQL

Core language for data extraction, joins, aggregations, and subqueries.

## Instagram Database

Real-world dataset with users, photos, likes, tags, and follows tables.

## PowerPoint / PDF

Final reporting format with query screenshots and insight summaries.

# Loyal Users & Inactive Accounts

## Task 1 – Loyal User Reward

Identify the **5 oldest users** by sorting on `created_at`. These early adopters are prime candidates for loyalty rewards.

📋 Top result: **Darby\_Herzog** — joined May 6, 2016

## Task 2 – Inactive User Engagement

Use a **LEFT JOIN** on `photos` and filter `WHERE photos.id IS NULL` to surface users who registered but never posted.

📋 **16 users** found with zero posts — re-engagement campaign targets identified.



# Contest Winner: Most-Liked Photo

By joining `photos`, `likes`, and `users`, then grouping by photo and ordering by like count, we identified the single most-engaged post on the platform.

48

Total Likes

Highest single-photo engagement in the dataset

145

Photo ID

The winning post identifier

52

User ID

Zack\_Kemmer93 — top content creator



# Hashtag Trends & Ad Timing

## Task 4 – Top 5 Hashtags

Grouping `photo_tags` by tag name reveals what content resonates most. **#smile** leads with 59 uses, followed by **#beach** (42) and **#party** (39).

## Task 5 – Best Day to Launch Ads

Aggregating registrations by `DAYNAME(created_at)` shows **Thursday and Sunday** tie for peak sign-ups (16 each), while Saturday is the slowest (12).

- 📌 Schedule high-priority campaigns on **Thursday or Sunday** for maximum reach.





## INVESTOR METRICS · TASK 1

# User Engagement: Posts Per User

A simple subquery divides total photos by total users, producing a clean platform-wide engagement baseline.

# 2.57

Avg Posts / User

Each user posts roughly 2-3 photos on average — a key benchmark for product health.

This metric helps investors gauge content depth and user investment in the platform over time.

# Bot & Fake Account Detection

Users who liked **every single photo** on the platform exhibit statistically improbable behavior. A `HAVING` clause filters for accounts where `total_likes = COUNT(*) FROM photos`.

⚠️ **15 suspicious accounts** detected — each with exactly 257 likes, matching the total photo count.





# Project Approach



The analysis leveraged five core SQL techniques: **JOIN** for combining tables, **GROUP BY** for aggregating results, **ORDER BY** for ranking, **aggregate functions** for counts and averages, and **subqueries** for comparative filtering.

# Key Takeaways & Impact

## Marketing Wins

Loyal users identified for rewards, 16 inactive accounts flagged for re-engagement, top hashtags mapped, and optimal ad days confirmed.

## Investor Signals

2.57 avg posts/user establishes an engagement baseline. 15 bot accounts flagged. Top creator (Zack\_Kemmer93) identified.

## Skills Demonstrated

Real-world SQL mastery — joins, aggregations, subqueries — applied to derive actionable business analytics from raw data.

